Alok Dadhich

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Education

Vishwakarma Institute of Technology **2022 – 2026**
Bachelor of Technology in Mechanical Engineering *Pune, India | CGPA: 7.90/10*

Experience

Indian Space Research Organisation (SAC, ISRO) **Mar. 2026 – Aug. 2026**
Research Intern *Ahmedabad*

- Designed the conceptual architecture and software for a low-cost **EOT crane-based positioning platform** for large-aperture antenna characterization, retrofitting an existing crane with **encoder-based sensing** and **RS-485/MEMOBUS** control.
- Built a **browser-based prototype** for coordinate management, raster scan planning, machine-limit enforcement, and operator interaction, validated on simulated data; authored a technical paper on the system.
- Developed a **MATLAB-based satellite antenna coverage prediction tool** using **WGS-84 ellipsoidal Earth modeling**, validated against industry-standard antenna analysis software on real **NVS-01 (GSLV-F12)** mission pattern data, achieving agreement within **0.01 dB**.

JSW Greentech **Aug. 2025 – Dec. 2025**
Body and Trim Design Intern *Pune*

- Created and modified **CAD models** for mechanical subsystems using AutoCAD and Creo; assisted in prototype validation, tolerance analysis, and BOM documentation.
- Collaborated with manufacturing and quality teams to improve component manufacturability and reduce rework.

Projects

Solar Still with Internal Condenser | *Dassault Systems Sponsored*

- Designed and fabricated a GI-body single-slope solar still (25° glass angle, epoxy-coated absorber) with **EPE insulation** and an analytically-sized **copper internal condenser** (9 tubes, ∼7.5 m length); ran controlled experiments on condenser mass flow rate (optimum **0.3 LPM**) and **PCM-based thermal storage**, achieving **34% energy efficiency** and a **25% increase** in net freshwater yield when combined.

3D-Printed Engine Prototype for Spark Advance Demonstration

- Fabricated a single-cylinder IC engine using 3D printing with Arduino Uno (C++) and rotary encoder feedback to study spark advance behavior.

Automatic Non-Electric Solar Tracker

- Designed and fabricated a passive, motor-free solar tracker driven by **R134a refrigerant**: differential heating of two canisters shifts vapor pressure and refrigerant mass, tilting the panel frame to follow the sun with no motors, sensors, or electronics; modeled the assembly in **Fusion 360** and built the frame in stainless steel/mild steel, validating **over 20% higher** power output than a fixed panel via logged voltage/current/power data.

CNC Coding and Robotic Application

- Designed and built a miniature **3-axis CNC machine** on an **Arduino Uno** with three **28BYJ-48 stepper motors** and **ULN2003 drivers** running **GRBL firmware** for G-code motion control; built the **Inkscape → JScut → UGS** software pipeline for vector-to-G-code conversion and 3D-printed the structural frame, achieving accurate, repeatable reproduction of lines, curves, and geometric patterns.

Technical Skills

Programming: C++, Python, MATLAB
Design & CAD: SolidWorks, AutoCAD, Creo, Fusion 360
Simulation: Ansys Fluent, CFD, FEA, MATLAB
Mechatronics: Arduino (C++), GRBL, Stepper Motor Control, CNC Integration
AI / Automation: Python, YOLOv8, OpenCV, Streamlit, Computer Vision

Leadership & Extracurricular

- **Under-Secretary-General, VIT Pune MUN** (Jul 2024 – Jul 2025) – Directed outreach campaigns and improved delegate participation by 10%.
- **Website Head, Vishwaconclave 2024** (Jan 2024 – Apr 2024) – Led web development for 2,500+ participants, improving registration efficiency.
- **Coordinator, Social Welfare and Development** (Jul 2023 – May 2024) – Organized the Go Green Drive with Aanandwan Foundation promoting sustainability awareness.

Achievements

- **Semi-Finalist (Top 50 Teams), DevStudio 2026 by Logitech** – solely designed and built **CADFlow**, a hardware-driven AutoCAD automation plugin, awarded Rs 60,000 worth of hardware. devpost.com/software/cadflow
- **High Commendation Award, SSI MUN 2024** (National-level)